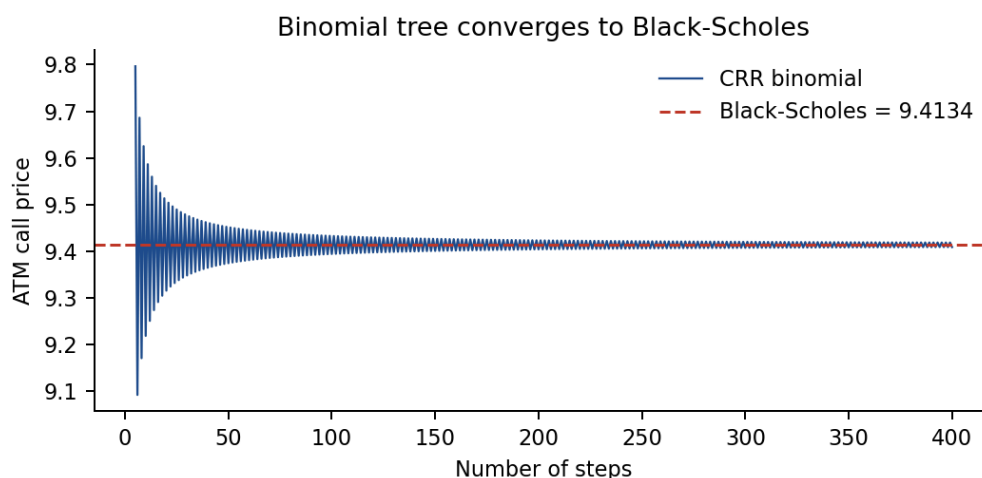


Options Lab

Pricing, Greeks & hedging from scratch in Python — does the theory survive discrete trading? · Eliote Demaret · EDHEC Business School

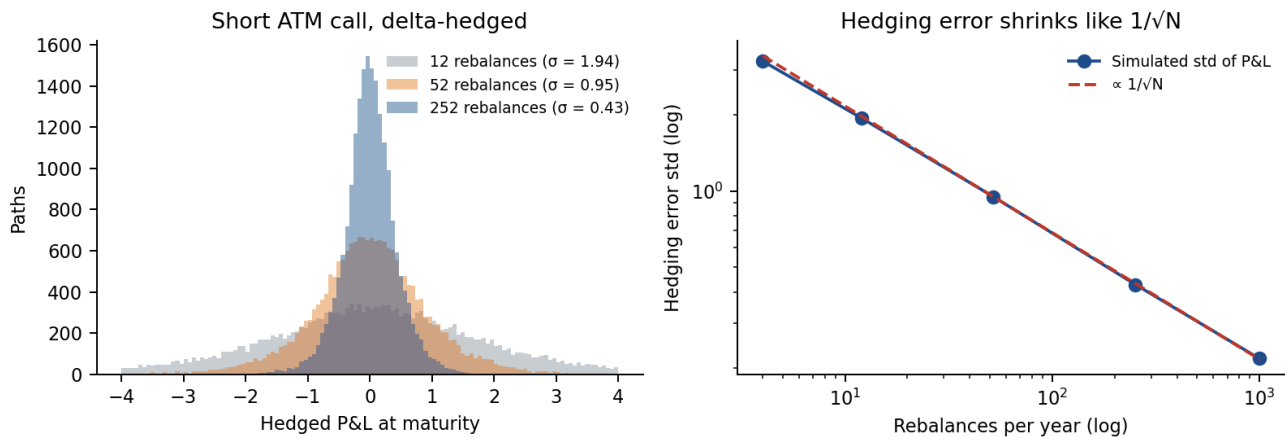
Summary. I built a Python library that implements Black-Scholes-Merton, a CRR binomial tree (European & American), Monte Carlo, implied volatility and analytical Greeks, all validated by 11 unit tests. I then used it to test five results from derivatives theory by simulation: (1) convergence of the tree to BSM, (2) how the hedging error depends on rebalancing frequency, (3) delta-hedged P&L as a bet on realised vs implied vol, (4) the minimum-variance futures cross-hedge, (5) how crash-type jumps create the equity volatility skew.

1. Binomial tree vs Black-Scholes



ATM 1-year call ($S = K = 100$, $r = 3\%$, $\sigma = 20\%$): BSM = 9.4134. The CRR tree oscillates between odd and even step counts and converges: error -0.0396 at 50 steps, -0.0050 at 400. On the same tree, an American put with $K = 110$ is worth 12.73 vs 12.04 for the European put. That 0.68 gap is the value of early exercise. For a call on a non-dividend stock the gap is exactly zero (tested).

2. Discrete delta hedging: hedging error vs rebalancing frequency

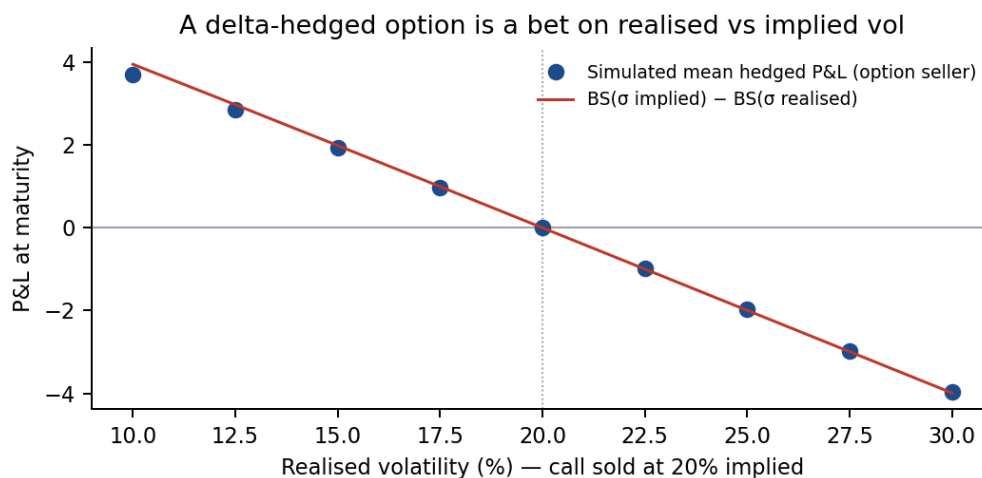


Sell one ATM call at the fair BSM price and delta-hedge along 20,000 simulated GBM paths (real-world drift 8%, vol = implied = 20%). With continuous hedging the P&L would be exactly zero. With discrete hedging the mean stays ≈ 0 but the dispersion depends on how often we rebalance:

Rebalances / year	4	12	52	252	1000
Std of hedged P&L	3.27	1.94	0.95	0.43	0.22
% of option premium	34.7%	20.7%	10.1%	4.5%	2.3%

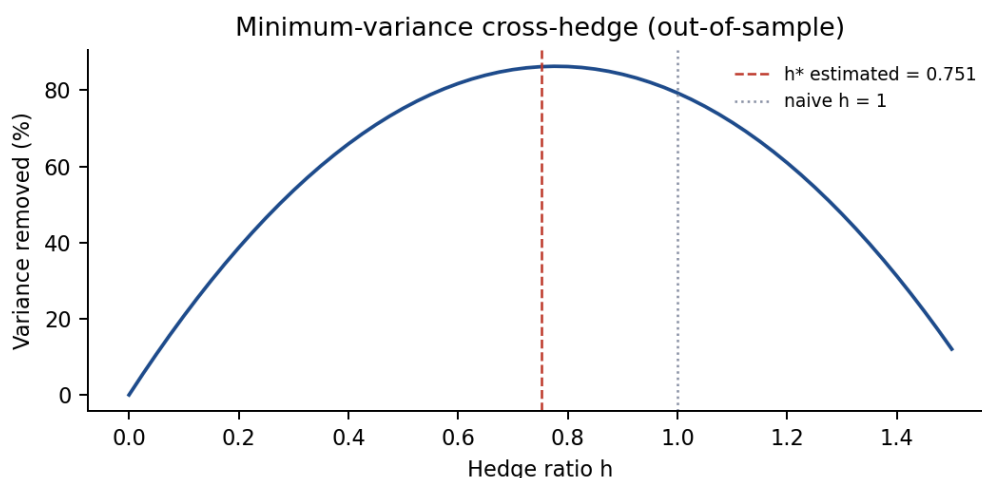
Result: the hedging error scales like $1/\sqrt{N}$. Going from weekly to daily hedging cuts the risk by about $2.2\times$ ($\approx \sqrt{(252/52)}$). The real-world drift doesn't matter, which is the replication argument behind risk-neutral pricing.

3. A delta-hedged option is a bet on realised vs implied volatility



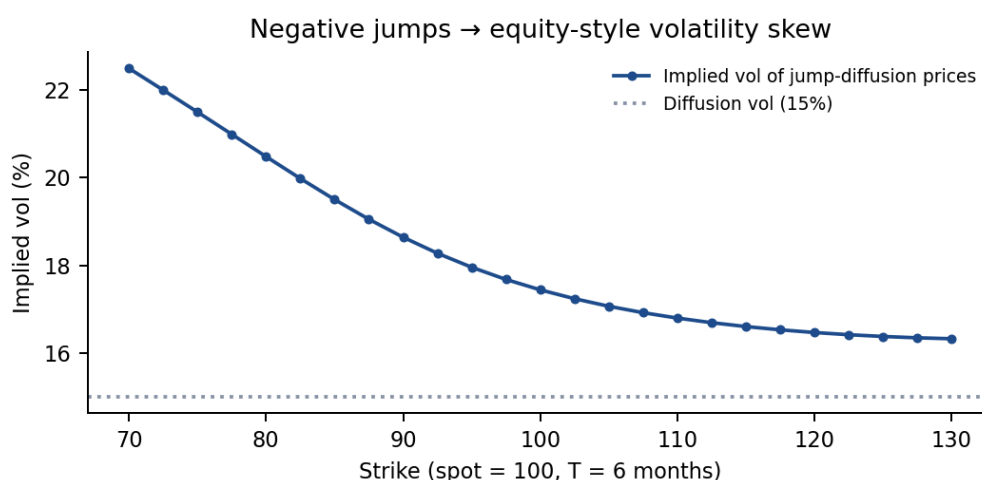
The call is sold at 20% implied vol and hedged daily with the 20% delta, while the stock actually realises between 10% and 30%. The seller's mean P&L tracks $BS(\sigma_{\text{implied}}) - BS(\sigma_{\text{real}})$: +1.93 if realised = 15%, -1.97 if realised = 25%. This is the gamma/theta trade-off: the short-gamma seller collects theta and loses $\frac{1}{2}\Gamma(\Delta S)^2$ on every move. That's why market makers who sell options are short volatility.

4. Minimum-variance futures cross-hedge



Set-up from Hull's airline example: jet fuel hedged with heating-oil futures ($\sigma_S = 2.63\%$, $\sigma_F = 3.13\%$, $\rho = 0.928$ monthly). Estimating $h^* = \rho\sigma_S/\sigma_F$ on 10 years of simulated monthly data gives 0.751 (true value 0.780). That means about 35 contracts with tailing for 2m gallons. Out-of-sample it removes 86% of the variance, against 79% for a naive one-for-one hedge. The residual is basis risk ($\approx 1 - \rho^2$). Beta hedging: moving a €10m portfolio from $\beta = 1.2$ to 0.6 with index futures at $5,000 \times €10$ means selling 120 contracts.

5. Where does the volatility skew come from?



I priced calls with a Merton jump-diffusion (15% diffusion vol, on average one -10% jump every 2 years) and inverted each price through Black-Scholes. Implied vol comes out at 22.5% at $K = 70$, 17.4% ATM and 16.3% at $K = 130$. Crash risk fattens the left tail, which makes OTM puts expensive relative to BSM and produces the downward-sloping equity skew seen in index option markets.

Limitations & next steps

Constant rates and volatility, no transaction costs. Next steps: add bid-ask costs to show the trade-off between hedging error and cost, calibrate to a real option chain (Euro Stoxx 50 / S&P 500), and backtest a delta-hedged straddle on historical data. Code, tests and figures are fully reproducible (pytest, python scripts/run_experiments.py).